IN SILICO MULTISCALE MODELING OF ENDOTHELIAL CELL MECHANOBIOLOGY IN A TUMOR MICROENVIRONMENT

BY

MUKTI SUBHASH CHOWKWALE

BE, University of Pune, 2015

THESIS

Submitted in partial fulfillment of the requirements for
the degree of Master of Science in Biomedical Engineering
in the Graduate School of
Binghamton University
State University of New York
2019

Accepted in partial fulfillment of the requirements for
the degree of Master of Science in Biomedical Engineering
in the Graduate School of
Binghamton University
State University of New York
2019

April 16, 2019

Dr. Gretchen Mahler, Chair and Faculty Advisor
Department of Biomedical Engineering, Binghamton University

Dr. Peter Huang, Faculty Advisor
Department of Mechanical Engineering, Binghamton University

Dr. Bruce Murray, Faculty Advisor
Department of Mechanical Engineering, Binghamton University

Abstract

Endothelial to mesenchymal transformation (EndMT) is a process in which endothelial cells gain a mesenchymal-like phenotype in response to mechanobiological signals that results in the remodeling or repair of underlying tissue. While initially associated with embryonic development, this process has since been shown to occur in adult tissue remodeling including wound healing, fibrosis, and cancer. In an attempt to understand the role of EndMT in cancer progression and metastasis, we present a multiscale, three-dimensional, in silico model which couples tissue-level, cellular-level, and molecular-level phenomena in the tumor microenvironment to mimic in vitro tissue models. This includes extracellular matrix remodeling, cell migration and proliferation, and signaling pathways that drive cellular transformation. The effects of varying substrate stiffness on EndMT and the effects of EndMT-derived fibroblasts on tumor cell migration and proliferation were simulated. Lastly, the model was used to explore the therapeutic role of EndMT inhibition in cancer treatment.

Acknowledgements

I would like to thank my thesis committee for their support and guidance over the past two years. Dr. Mahler has been a great advisor and boss, and I would like to thank her for patiently bearing with me frequently bursting into her office with a lot of questions and problems. She has been very kind and patient in sharing her expertise and knowledge with me. Dr. Huang's insight during our project meetings has made me more critical of my work, and I sincerely thank him for our invaluable discussions during the research meetings. I am profoundly grateful to Dr. Murray for being a mentor and friend. His patience, kindness, and guidance has made my time at Binghamton a thoughtful and rewarding journey. The constant support and faith of my committee in me has helped me believe in myself and shaped me to be a independent and thoughtful researcher.

I'm grateful to have been a part of the Mahler research group, and I would especially like to thank the EndMT group for their help and suggestions. Jonathan and Bridget put a lot of time and effort into teaching me cell culture and imaging techniques, and supervising me while I performed experiments.

I have enjoyed my time in Binghamton because of the people I have met here and spent time with. It isn't possible to list everyone, but a special thanks to Aagya,

Janani, Ethan, Azam, Stephanie, and Adia for always being there for me. I would like to finish by thanking my friends and family from back home, for without them I would not be the person I am today.

Contents

List of Tables

List of Figures

List of Abbreviations

α-SMA alpha Smooth Muscle Actin

ABM Agent-based modeling

CA Cellular automata

CAF Cancer-associated Fibroblasts

COPASI COmplex PAthway SImulator

CPM Cellular Potts model

ECM Extracellular Matrix

EMT Epithelial to Mesenchymal Transformation

EndMT Endothelial to Mesenchymal Transformation

ERK Extracellular signaling Receptor Kinase

FAK Focal Adhesion Kinase

FSP-1 Fibroblast Specific Protein 1

IL-1 Interleukin 1

MCS Monte Carlo Step

MDEs Matrix Degrading Enzymes

MMPs Matrix Metalloproteinases

ODEs Ordinary Differential Equations

PECAM-1 Platelet Endothelial Cell Adhesion Marker 1

PI3K Phosphoinositide 3 Kinase

pSmad2 Phosphorlyated Smad2

pSmad3 Phosphorlyated Smad3

R-Smads Receptor-activated Smads

RoG Radius of Gyration

ROS Reactive Oxygen Species

SBML Systems Biology Markup Language

SIS3 Specific Inhibitor of Smad3

TGF-β Transforming Growth Factor beta

TNF-α Tumor Necrosis Factor alpha

uPAs Urokinase Plasminogen Activators

VE-Cadherin Vascular Endothelial Cadherin

1 Introduction

1.1 Modeling in Multicellular Biological Systems

The modeling of multicellular biological systems incorporates complex interactions among cells and their environment to provide a holistic perspective of the studied system [1]. This can include dynamic cell behavior such as cell growth and proliferation, migration, and apoptosis, as well as environmental factors such as the extracellular matrix, diffusing chemicals, and forces which influence cell behavior. While *in vitro* experiments provide insight into specific, limited aspects of a biological system, *in silico* models provide a means of incorporating biological, chemical and physical effects in the study of a complex phenomenon to provide a more comprehensive picture. The conclusions from such models can be used to form new hypotheses that can be tested with laboratory experiments. Furthermore, these models allow subjecting the *in silico* biological environment to conditions which are difficult to reproduce in *in vitro* or *in vivo* settings [2]. Modeling approaches are often grouped into two broad categories, continuum models and discrete (or cell-based) models. Simulation of complex biological phenomena often requires a blending of the two approaches, yielding what are called hybrid models.

1.1.1 Continuum Models

Continuum models typically deal with phenomena that occur on somewhat larger temporal and spatial scales [3], for example, at the tissue level. In continuum models, cellular behavior is represented in terms of cell densities and chemical constituents and macromolecules are treated in terms of concentration levels. These quantities are field variables which depend on space and time and are subject to mechanobiological principles that are represented in terms of partial differential equations [3]–[5]. The continuum approach is well-suited for mesoscale phenomena with large cell populations. They are not able to describe individual cell behavior, such as the cell-specific behavior that arises from endothelial to mesenchymal transformation (EndMT). Moreover, some of the assumptions made while deriving continuum models are difficult to relate to individual cell behavior [6]. They are generally easier to implement owing to the well established means of solving the governing differential equations numerically.

1.1.2 Discrete Models

In contrast, discrete cell models allow for simulating individual cell behavior and interactions among them. Cellular and subcellular experimental data can be incorporated in formulating discrete models, hence enabling the simulation of emergent system behavior. However, modeling large cell populations using this method can be computationally intensive. Hybrid discrete-continuum models are used to overcome some of the drawbacks of cell-based models.

There are a number of approaches used in discrete cell modeling. They can be broadly categorized as based on biophysical principles or rule based. Two widely used methods in the latter category are cellular automata (CA) and agent-based modeling (ABM). CA models have four fundamental components: a discrete lattice, a finite set of cellular states, a finite set of neighboring cells, and rules for cell state transition [1]. Inferences regarding the mechanisms for cell behavior are made based on these components [7]. In contrast, ABMs involve decision-making components called agents, which follow a set of rules to interact with the environment. The simulated mechanisms which result from the sets of rules for agents generate the emergent system behavior. Physically based approaches treat cells as either rigid or elastic particles that move and interact based on mechanobiological fundamentals.

1.1.3 Lattice Systems

A lattice is an n-dimensional closed grid, characterized by periodic or fixed boundary conditions in each dimension [8]. The size of the lattice elements is decided based on the application of the model; the distance between adjacent elements can be made comparable to biological cells, multiple cells, or subcompartments of the cells [1]. The primary advantage of lattice-based modeling is that it is well-suited for describing interconnected processes at different spatial levels, i.e., microscopic (molecular interactions), mesoscopic (cell behavior, cell-cell interactions, and cell-matrix interactions), and macroscopic (tissue and organ properties) scales [9]. Contrary to lattice-based systems, lattice-free models allow the cells to be at any location in the simulated domain. Equations representing the forces which act on the cells are solved to determine the positions of the cells. This allows a mechanistic description of cell-cell and cell-substrate interactions without introducing artificial constraints [10]. The choice of on-lattice or off-lattice models depends on the level of biological detail that is to be simulated; on-lattice models generally require a simple cell geometry but permit simulating large cell populations, whereas off-lattice models have advantages for fully deformable cell shapes but typically requires smaller cell populations due to computational costs [11].

1.1.4 Agent-based Modeling

Agent-based modeling is a computational modeling methodology that is rule-based, discrete-event based, and discrete-time based. It allows incorporation of spatial scales and stochasticity, utilizes parallelism, creates a modular structure, and permits model creation in the absence of absolute biological knowledge [7]. Biomedical applications of ABMs include intracellular modeling [4], [12], cell-level tissue modeling [13]–[15], and multiscale approaches [16]–[18]. The three types of ABMs used for describing cell behavior - cell-center, cell-vertex, and Cellular Potts models - are shown in Figure 1.1 [19].

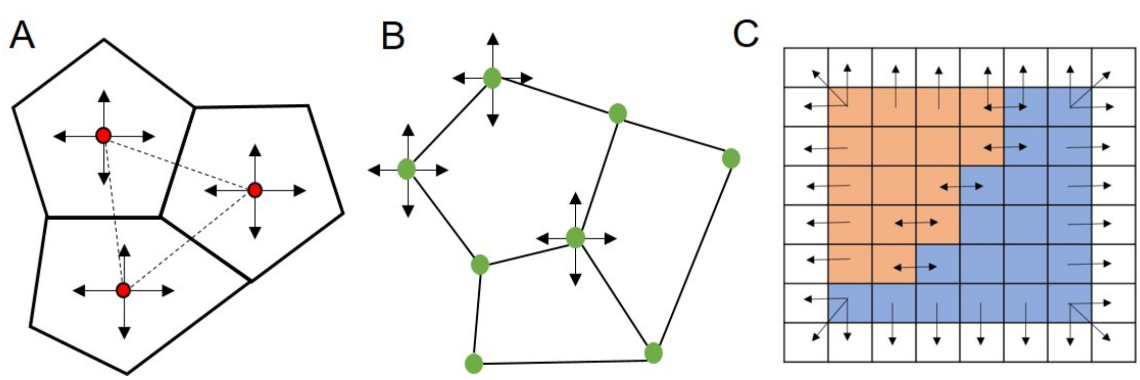

Figure 1.1: (A) Cell-center, (B) cell-vertex, and (C) Cellular Potts models.

Cell-Center Models

Cell-center models are used to represent aggregates of biological cells, where the spatial position of each cell is defined by the cell center [20]. Cell behavior is modeled as a function of the cell center. For example, cell proliferation is the addition of new cell centers, apoptosis is the deletion of an existing cell center, cell growth is the increase in deviation from the cell centers, and cell migration is movement of the cell center. Interactions between the centers can represent forces that act upon the cells [19]. As a result of these modeled interactions, simulating differential cell-cell and cell-substrate adhesion is straightforward. However, a disadvantage of this approach arises from a geometric limitation known as Delaunay triangulation, where lines from the vertices of a polygon do not meet at one center. This limitation does not allow the shapes of the cells to change smoothly [21].

Cell-Vertex Models

In cell-vertex models, cells are represented as adjacent polygons. As the edges are explicitly defined, the effect of mechanical forces such as cell plasticity and elasticity can be modeled in a straightforward manner [20]. Forces on the vertices and free energy functions which balance the forces between cells can also be modeled using this approach [19]. Consequently, cell-vertex models can describe cell shape changes more realistically than cell-center models [21]. However, the modeling of cell-substrate adhesion is complicated, as the force terms include contributions from surrounding cells.

Cellular Potts Models

Cellular Potts models are lattice-based simulations, where cells are represented as multiple sites in a lattice system. This method allows for the modeling of differential adhesion, properties of cell morphology, and cell migration [19]. The Cellular Potts model is described thoroughly in Section 1.2.

1.1.5 Agent-based Modeling Software: Netlogo vs CompuCell3D

Netlogo [22] is an agent-based modeling environment used to for complex systems. It allows for rule-based modeling by executing procedures attached to agents. CompuCell3D [23] is a software implementation of the Cellular Potts Model, which has a typical application in the modeling of soft tissue with motile cells. These two software were evaluated for determining which would be well-suited for this study. The comparison between the two is summed up in Table 1.1.

1.2 Cellular Potts Model

The Cellular Potts model, also known as the Glazier-Graner-Hogeweg Model, was derived from the large-Q Potts model, which was used for simulations of diffusive grain growth. This method was driven by surface energy and could correctly simulate observed topological changes in cellular patterns in metals and soap froths [24]. The large-Q Potts model is a generalization of the Potts model where the limit $q \to \infty$ [25]. The q-value refers to the possible states at a vertex in the lattice of the q-state Potts model, which itself was a generalization of the Ising model. The Ising model

	Netlogo	CompuCell3D
Resource Requirement	High, as it is written in Java	Not as high as Netlogo, since it is written in C++/Python and uses XML
Time Requirement	Time-intensive due to Java and lack of 3D support	Much faster for a model comparable to Netlogo
Complexity	Simple, as it is rule-based	Math-intensive
3D Modeling	Limited support	Well developed and documented platform
Continuum Model	Can be modeled	Can be modeled
Signaling Model	Can be modeled, and can integrate with other software for faster simulation	Can be modeled using SBML
Forces acting on cells	Effects like displacement can be modeled, but cell deformation cannot	Effects can be modeled as a change in pressure and/or volume
Integration with other software	Possible, has been used with MATLAB, FEBio, and Computational Fluid Dynamics software [13], [16]–[18]	Complicated

Table 1.1: Comparison between the features of Netlogo and CompuCell3D

was developed to describe the overall ferromagnetic properties of a system as a result of the interactions of individual magnetic spins in the system [26]. The motivation behind creating the Cellular Potts model was to mimic cell sorting that occurred in embryonic cells. It had been observed that embryonic cells of two different types when dissociated, randomly mixed, and reaggregated, spontaneously sorted to reestablish homogeneous tissue. This behavior was observed in both 2D and 3D tissue models. This sorting was based on the spatial rearrangement of the cell positions, and experiments had shown that differences in cell adhesion for different cell types resulted in the final, sorted state which had the global minimum of overall surface energy. Thus, the Cellular Potts model (CPM) is based on the differential adhesion hypothesis which allows the simulation of cell movement and rearrangement depending on the minimization of cell-cell surface interaction energy [4]. The differences in the cell adhesion energies cause quantifiable tissue surface tension, which any biological process aims to minimize. This concept forms the core of the Cellular Potts model, known as the effective energy function or the Hamiltonian function [26]. The energy function is considered "effective" as it represents internal cell responses in addition to external forces. The generalized cell is modeled as a deformable object, whose shape is determined by the internal and external forces acting upon it [27]. Consequently, a generalized cell can be used to represent a biological cell, an intracellular element, or a part of the extracellular matrix. The initial CPM [24] is described using a collection of N cells by defining N degenerate spins $\sigma(i,j) = 1, 2, 3, ..., N$ where (i,j) corresponds to a lattice site.

In Figure 1.2, each cell has the spatial location (i,j) and a domain index σ. The type of the domain is represented by $\tau(\sigma)$. In this case, the possible types are A or B. Adhesion between different cell types has the energy 1, while adhesion between similar cell types has the energy 0. The Hamiltonian function of the system is then modeled as follows:

$$H = \sum_{i,\ j,\ neighbors} (1 - \delta_{\sigma(i),\ \sigma(j)}), \tag{1.1}$$

where $\delta_{\sigma(i),\ \sigma(j)}$ is the Kronecker delta, which ensures that neighboring lattice sites of the same cell do not contribute to the overall energy of the system [27]. To determine

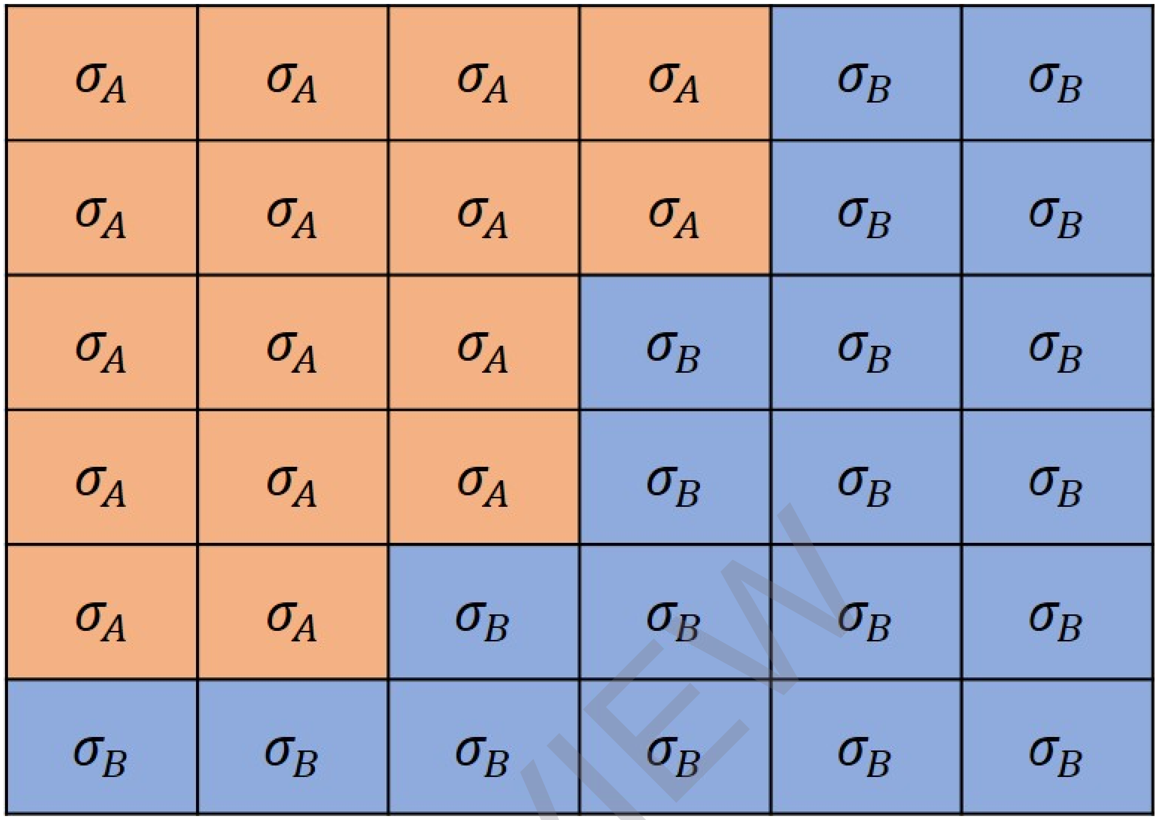

Figure 1.2: A 6 x 6 lattice described by the Cellular Potts Model. Each square has a spatial location (i, j) and a domain index σ.

the change of the cell spin, the Metropolis Monte Carlo algorithm is used. In each Monte Carlo step, a nearest-neighbor-order site is selected at random. The spin of the site is changed from σ to σ'. The Monte Carlo success probability of this change is calculated as,

$$P(\sigma(i,j) \to \sigma'(i,j)) = e^{-\Delta H/kT}, \tag{1.2}$$

where ΔH is the gain in system energy, T is the temperature, and k is the Boltzmann constant. The temperature does not represent the actual physical temperature, but instead describes the amplitude of cell-membrane fluctuations [23]. This is repeated for N times per cell, and the spin change that results in the minimum energy change is selected. Above the critical temperature T_c, cells begin to dissociate, while below this temperature the cells cluster together. At $T = 0$, cells grow at relaxation, as the pattern minimizes its surface area. An extended model was developed, which uses a second term to sum up all the constitutive relations of the cell as a function of their area or volume. This allowed including cellular behavior such as membrane elasticity, cytoskeletal properties, etc [28]. Further extensions of the model formed the basis of CompuCell3D. Advantages of CPM are the simplicity of implementation and its speed compared to other algorithms. Moreover, the model uses the position and diffusion of the membrane to determine the cellular dynamics, which is similar to the behavior of biological cells wherein the motion is driven by relative boundary curvatures and relative contact energies. However, there are limitations of the CPM due to its lattice-based implementation. The lattice discretization and anisotropy can cause fluctuations in contact angles and boundary pinning. The finite temperature of the system can also cause thermal fluctuations in contact angles and boundary shapes. Furthermore, the assumptions of the model enforce cell isotropy, and so a polarized cell with varying membrane elasticity or surface chemistry cannot be created using CPM.

1.3 Modeling in Cancer

Mathematical and computational modeling of cancer enables the simulation of the interactions between cancer cells, normal cells and the surrounding environment. The linking of the molecular and cellular scale-events with tissue level phenomena allows studying the individual and combined effects of these events on tumor formation and